Prime-Power Divisibility of Fibonacci Hypercube Sums

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Abstract

We prove the d-dimensional prime-power divisibility conjecture stated alongside the cubic Fibonacci-array sequence OEIS A129762: for every positive dimension d and every prime $p \equiv \pm 1 \pmod 5$, the sum over the $(p-1)^d$ array is divisible by p^d. This includes the stated cubic p^3 observation and all primes p dividing F_{p-1}. The proof uses the splitting of $x^2 - x - 1$ modulo p and Binet's formula.

OEIS assertion. A129762 is $a(n) = \sum_{1 \le i,j,k \le n} F_{i+j+k-2}$, with $F_0 = 0$, $F_1 = 1$. Its comments state $p^3 \mid a(p-1)$ for primes $p \equiv 1, 4 \pmod 5$ and conjecture the analogous p^d divisibility for d-dimensional arrays whenever $p \mid F_{p-1}$.

Theorem 1. *Let $d \ge 1$ and let p be a prime with $p \equiv 1$ or $4 \pmod 5$. Put*

$$S_d(n) = \sum_{i_1,\dots,i_d=1}^{n} F_{i_1+\cdots+i_d-d+1}.$$

Then $p^d \mid S_d(p-1)$. Moreover, every prime p with $p \mid F_{p-1}$ belongs to these two residue classes modulo 5, so the OEIS d-dimensional conjecture follows.

Proof. Set $R = \mathbb{Z}[\alpha]$ with $\alpha = (1+\sqrt{5})/2$ and $\beta = (1-\sqrt{5})/2 = 1 - \alpha$. Both are algebraic integers, $\alpha\beta = -1$, and $\alpha - \beta = \sqrt{5}$. The prime $p \equiv \pm 1 \pmod 5$ is odd and differs from 5. Quadratic reciprocity gives $(5/p) = 1$, so $x^2 - x - 1$ has two distinct roots in $\mathbb{F}_p$. The quotient R/pR is consequently isomorphic to $\mathbb{F}_p \times \mathbb{F}_p$. The images of α and β in each factor are nonzero, since their product is -1. Fermat's theorem in each factor implies

$$\alpha^{p-1} - 1 \in pR, \qquad \beta^{p-1} - 1 \in pR.$$

Binet's formula and the finite geometric series, using $\alpha - 1 = \alpha^{-1}$ and $\beta - 1 = \beta^{-1}$, give the exact identity

$$S_d(n) = \frac{\alpha^{d+1}(\alpha^n - 1)^d - \beta^{d+1}(\beta^n - 1)^d}{\alpha - \beta}.$$

For $n = p - 1$ the numerator belongs to $p^d R$. As $\alpha - \beta = \sqrt{5}$ is a unit modulo p (its square is 5), it follows that $S_d(p-1) \in p^d R$. But $S_d(p-1)$ is an ordinary integer and $p^d R \cap \mathbb{Z} = p^d \mathbb{Z}$ (the basis $1, \alpha$ is integral). Hence $p^d \mid S_d(p-1)$.

For completeness, suppose $p \mid F_{p-1}$. The primes 2 and 5 do not satisfy this ($F_1 = 1$, $F_4 = 3$). If $(5/p) = -1$, Frobenius interchanges the two roots in $R/pR \cong \mathbb{F}_{p^2}$: $\alpha^p = \beta$ and $\beta^p = \alpha$. Thus $\alpha^{p+1} = \beta^{p+1} = -1$, giving $F_{p+1} \equiv 0 \pmod p$. Together with $F_{p-1} \equiv 0 \pmod p$ this implies $F_p = F_{p+1} - F_{p-1} \equiv 0 \pmod p$, and then $F_{p-2} = F_p - F_{p-1} \equiv 0 \pmod p$. Repeating backwards forces $F_1 \equiv 0 \pmod p$, impossible. Thus $(5/p) = 1$, equivalent to $p \equiv \pm 1 \pmod 5$. □

Scope and priority. This is a direct proof of the specific multidimensional divisibility statement recorded as a conjecture in A129762. It is a classical algebraic-number argument, and no claim of novelty or independent priority is made. It does not address primality of individual array sums.

References

[1] OEIS Foundation, *A129762: Sum of Fibonacci entries in a cubic array*, https://oeis.org/A129762 (accessed November 14, 2026).

[2] OEIS Foundation, *A000045: Fibonacci numbers*, https://oeis.org/A000045 (accessed November 14, 2026).

Authorship and verification. Primary author (AI): ChatGPT (OpenAI). This self-contained mathematical argument was prepared with automated arithmetic checks and source comparison. It has not received independent human peer review; no originality or priority is claimed.